Abinaya Meenatchisundharam

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PROFESSIONAL SUMMARY

Software Engineer with 1+ year of professional experience building Go services, backend APIs, and Linux networking systems, complemented by 3 years of part-time C/C++ embedded and IoT development alongside a B.E. in Electronics and Communication Engineering. Experienced in concurrent systems, Linux network configuration, REST APIs, integration testing, and hardware-software debugging. Promoted within 11 months, mentored new engineers, and trained 30+ students in embedded systems and IoT.

TECHNICAL SKILLS

Languages: Go, C, C++, Python, Bash
Backend: REST APIs, concurrency, goroutines, state machines, PostgreSQL, FastAPI, JWT/JWKS, RBAC
Systems & Networking: Linux, netlink, namespaces, systemd, TCP/IP, VLAN, VXLAN, NAT, routing, QoS, OSPF/BFD
Embedded: Arduino, ESP32/ESP8266, Raspberry Pi, UART, I2C, MQTT, Wi-Fi, BLE, QMI/MBIM/AT
Testing & Infrastructure: Go testing, race detection, mocks, integration testing, Git, tcpdump, Wireshark, iperf3, KVM/libvirt, CloudStack

PROFESSIONAL EXPERIENCE

Junior Software Developer **Aug 2025 – Sep 2026**
Karios HCI Pvt. Ltd. – promoted from Software Developer Trainee, Jul 2026 *Remote, Bengaluru*

- Engineered a concurrent Go uplink-failover service that probes fiber and LTE links in parallel, scores latency, loss and jitter, and switches routes on data-plane failure; validated recovery within 105 seconds and added cooldown logic to prevent route flapping during repeated modem disconnects.
- Brought up Quectel EC25 and Sierra EM7411 LTE modems on Linux over QMI, MBIM and AT; traced silent failover to a missing gateway host route and fixed it with the route plus a readiness gate; wrote an RSRP/SINR band-selection engine in Go with anti-flap logic.
- Implemented netlink-based Go APIs for VLAN, bridge and interface configuration on a staged, asynchronous job lifecycle with ownership checks and rollback, plus per-interface QoS (HTB, AQM, CAKE) verified by reading the qdisc tree back from the kernel; replaced SSH-based provisioning with an authenticated cross-node API client.
- Traced a 40G throughput ceiling (∼23–24 Gbps in iperf3) to CPU softirq saturation in the Linux bridge and HTB path, ruling out the NIC, RSS and QoS through queue-tuning tests.
- Hardened multi-tenant networking on Apache CloudStack and PostgreSQL: idempotent provisioning, HTTP 409 conflict handling, rollback of partial allocations and ordered deletion with VM and IP guards, leaving zero orphaned resources in end-to-end tests.
- Delivered VXLAN network-ID allocation with a background Go reconciler syncing CloudStack to NetBox; fixed a silent sync failure and a tenant-visibility bug in CloudStack ACLs, then verified VXLAN, bridge and FDB state through to VM-to-VM ping.
- Added JWT/JWKS authentication to an edge-appliance API: key caching, RS256 and claim validation, route-level 401/403 permissions, and token revocation that fails closed when Redis is unavailable.
- Co-developed Go integration suites for network and VPC APIs; triaged 36 nightly failures into test, environment and backend causes, escalating backend defects instead of weakening assertions. Earlier authored suites for 4 platform services (3 merged).
- Ran a two-week production knowledge-transfer program for new engineers; built FastAPI/PostgreSQL device APIs with JWT and RBAC during onboarding.

IoT & Embedded Developer **Dec 2021 – Dec 2024**
KGXperience Innovation Cell, KGISL Institute of Technology (part-time, alongside B.E.) *Coimbatore, India*

- Led student teams building IoT prototypes in C/C++ on Arduino and ESP32/ESP8266 and Python on Raspberry Pi, integrating sensors over UART, I2C, Wi-Fi and BLE from design to live demo.
- Set up MQTT publish/subscribe telemetry with a Mosquitto broker and paho-mqtt during the internship. Ran 3 IoT and embedded bootcamps for 30+ students on microcontrollers, protocols and debugging; wrote ESP8266 demo firmware that joins Wi-Fi and calls a webhook to send email alerts.

PROJECTS

8-Bit Breadboard Computer **Jul 2023 – Jan 2024**
Team Lead | 74-series TTL logic, EEPROM, Arduino Nano, C++ *GitHub | Demo video*

- Built a working 8-bit computer from 74-series logic ICs (Ben Eater design): clock, program counter, A/B registers, ALU, 16-byte RAM, instruction register and EEPROM-based control logic.
- Programmed control microcode and a multiplexed 7-segment display driver into EEPROMs in C++ via an Arduino Nano; debugged clock-timing faults and breadboard signal loss.

EDUCATION

B.E. Electronics and Communication Engineering, KGISL Institute of Technology Nov 2021 – May 2025